KU LEUVEN

FACULTY OF ENGINEERING TECHNOLOGY

CAMPUS GROUP T

Lab Report for Complex Digital Design

Final Project

28/04/2024 (submission date)

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Academic year 2023-2024

1 SUMMARY

For our final project we implemented a Hybrid adder using Carry Select Adder module with Carry Look Ahead adder as its submodules. Our design improves speed performance, having a trade-off on the area. The maximum Adder_Width tolerated by our design is 64-bit with Worst Negative Slack equal to 1.56 ns. Therefore, the maximum number of cycles taken is 10.

2 TECHNICAL DESCRIPTION

To improve the performance of the multi-precision adder, we tackle the two main draw backs of the Ripple Carry Adder (RCA) as follows:

1- Implementing a Uniformed-sized carry select adder (USCA). This allows every Full Adder (FA) to output a predicted a result (sum and carry) without waiting for the previous FA C_{in}

2- Implementing a Carry Look Ahead adder (CLA) for the M-bit adder to lower the worst-case carry propagation delay between the single adder blocks.

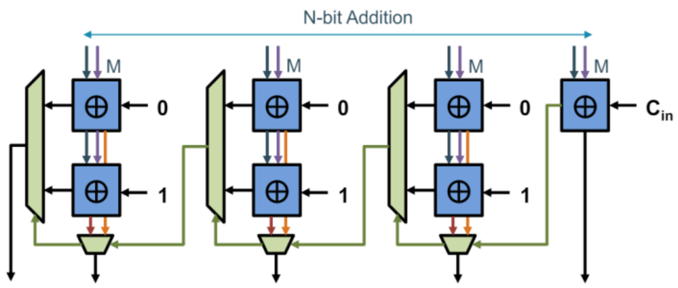

Figure 1: Uniformed-sized carry select adder.

The code logic for the design shown in Figure 1 is in UCSA.v source file. First, the M-bit RCA is used as a sub module for the CSA. This is replaced later by the CLA adder. The worst-case delay now goes through the first RCA block and the different multiplexers. With this approach we reduce the propagation delay by performing pre-computations with duplicated logic. The other source files such as: CSA_2bit.v, CSA_4bit.v were used as step-by-step approach.

We then create a hybrid adder. This is done in certain steps. The full adder (FA) with the Partial Full Adder (PFA). We then create the Carry Propagation logic block by expanding the expressions of all C_i to depend only on C_0, which is actually C_{in}. The number of PFA, M, is 8. This code logic is found on CLA.v source file. We then replace the M-Bit RCA with the CLA in the CSelA module. By doing this we have a high-speed performance improvement for our adder circuit. However, it has a significant area increase and signals with very large fan-outs. This is the trade off between speed and area for our strategy. Our final project adder module used on the mp_adder is shown in Figure 2. Testbenches for each Source File are created in Simulation Sources.

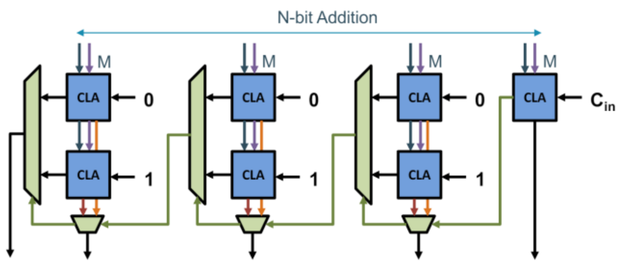

Figure 2: The Hybrid Adder, the final combinational adder for the project.

3 PERFORMANCE EVALUATION

The hybrid adder was chosen for the final project for its improvements in speed when larger integers are being added. Below is a summary of its performance from the post-synthesis timing report:

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	3,177 ns	Worst Hold Slack (WHS):	0,132 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	7502	Total Number of Endpoints:	7502	Total Number of Endpoints:	3218

All user specified timing constraints are met.

Figure 3: Hybrid adder timing report for Adder_Width = 32 bits and 8 PFA for each CLA

Figure 3 shows a worst negative slack (WNS) of 3,177 ns. The typical worst-case paths are summarized in a more detailed report below (Figure4):

Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock
Path 1	3.177	4	17	design_1_i/u...B_Q_reg[3]/C	design_1_i/u..._reg[509]/D	4.687	1.145	3.542	8.000	sys_clk_pin	sys_clk_pin
Path 2	3.177	4	17	design_1_i/u...B_Q_reg[3]/C	design_1_i/u..._reg[510]/D	4.687	1.145	3.542	8.000	sys_clk_pin	sys_clk_pin
Path 3	3.177	4	17	design_1_i/u...B_Q_reg[3]/C	design_1_i/u..._reg[511]/D	4.687	1.145	3.542	8.000	sys_clk_pin	sys_clk_pin
Path 4	3.414	4	17	design_1_i/u...B_Q_reg[3]/C	design_1_i/...gCout_reg/D	4.450	1.145	3.305	8.000	sys_clk_pin	sys_clk_pin
Path 5	3.481	3	20	design_1_i/u...Cnt_reg[2]/C	design_1_i/u...Res_reg[0]/R	3.782	1.013	2.769	8.000	sys_clk_pin	sys_clk_pin
Path 6	3.481	3	20	design_1_i/u...Cnt_reg[2]/C	design_1_i/u...Res_reg[1]/R	3.782	1.013	2.769	8.000	sys_clk_pin	sys_clk_pin
Path 7	3.481	3	20	design_1_i/u...Cnt_reg[2]/C	design_1_i/u...Res_reg[2]/R	3.782	1.013	2.769	8.000	sys_clk_pin	sys_clk_pin
Path 8	3.481	3	20	design_1_i/u...Cnt_reg[2]/C	design_1_i/u...Res_reg[3]/R	3.782	1.013	2.769	8.000	sys_clk_pin	sys_clk_pin
Path 9	3.481	3	20	design_1_i/u...Cnt_reg[2]/C	design_1_i/u...Res_reg[4]/R	3.782	1.013	2.769	8.000	sys_clk_pin	sys_clk_pin
Path 10	3.481	3	20	design_1_i/u...Cnt_reg[2]/C	design_1_i/u...Res_reg[5]/R	3.782	1.013	2.769	8.000	sys_clk_pin	sys_clk_pin

It is the worst delays are observed between the time the register inputs the 4th-bit of the already shifted, received operand from uart_rx as an input operand to the hybrid adder, and the time it is outputted as the last 3 bits of the results (indicated as regResult_reg[509] to regResult_reg[511]). This is expected, since the last bits of the result is a very high number and the combination of using a CLA and UCSA produces high fanouts.

This timing report is a great improvement compared to the two previous adder configurations used, the UCSA and the RCA. For the RCA, 32-bit integer additions would fail the time constraints, while the UCSA produces the following time report:

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	2,491 ns	Worst Hold Slack (WHS):	0,137 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1579	Total Number of Endpoints:	1579	Total Number of Endpoints:	1580

All user specified timing constraints are met.

Figure 4: UCSA timing report for Adder_Width = 32 bits and 8-bit RCA

A lower slack suggests a higher worst-case path is found, thus the difference between the actual timing constraint to be met is lower. This proves that the hybrid adder would indeed produce a faster processing.

However, when the amount of PFA or size of RCA is smaller, this effect is less obvious. Observe the following phenomena:

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	2,468 ns	Worst Hold Slack (WHS):	0,137 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1570	Total Number of Endpoints:	1570	Total Number of Endpoints:	1571

All user specified timing constraints are met.

Figure 5: UCSA timing report for Adder_Width = 32 bits and 4-bit RCA

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	2,040 ns	Worst Hold Slack (WHS):	0,132 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1554	Total Number of Endpoints:	1554	Total Number of Endpoints:	1555

All user specified timing constraints are met.

Figure 6: Hybrid adder timing report for Adder_Width = 32 bits and 4 PFA for each CLA

Figure 5 and Figure 6 illustrates a small difference between the two WNS. The reason for this is the scaling impact. When more complicated modules and logic are being used, the inherent delays may be more potent when the amount of adders used are still small. However, when the scaling is large enough, the impact is not prevalent.

It is also worth mentioning the timing report of the hybrid adder when a 64-bit Adder_Width is used as follows:

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	1,560 ns	Worst Hold Slack (WHS):	0,132 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1570	Total Number of Endpoints:	1570	Total Number of Endpoints:	1571

All user specified timing constraints are met.

Figure 7: Hybrid adder timing report for Adder_Width = 64 bits and 8 PFA for each CLA

A decrease in WNS is observed due to the increase of Adder_Width that created more delays (Figure 7).

Finally, the utilization report is shown below (Figure 8):

Name	Slice LUTs (53200)	Slice Registers (106400)	Bonded IOB (125)	BUFGCTRL (32)
N mp_adder	1198	1570	1541	1

Figure 8: Hybrid adder utilization report for Adder_Width = 32 bits and 8 PFA for each CLA

4 COMPARISON WITH LAB#3

A comparison of the obtained performance metrics with respect to the original ones from Lab #3, that is, when MP ADDER uses the ripple_carry_adder_Nb with ADDER_WIDTH = 16 is given as follows.

Speed Comparison

We can see in Figure 3 that the Hybrid Adder reports a bigger Worst Negative Slack than the Ripple Carry Adder in Figure. This means that now the Total delay of the most critical path is smaller, making the speed performance increase.

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	3,555 ns	Worst Hold Slack (WHS):	0,137 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1575	Total Number of Endpoints:	1575	Total Number of Endpoints:	1576

All user specified timing constraints are met.

Figure 9: The Post-Synthesis report of the hybrid adder tested with Adder_Width = 16.

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	2,533 ns	Worst Hold Slack (WHS):	0,137 ns	Worst Pulse Width Slack (WPWS):	3,500 ns
Total Negative Slack (TNS):	0,000 ns	Total Hold Slack (THS):	0,000 ns	Total Pulse Width Negative Slack (TPWS):	0,000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1573	Total Number of Endpoints:	1573	Total Number of Endpoints:	1574

All user specified timing constraints are met.

Figure 10: The Post-Synthesis report of the ripple carry adder from LAB#3 tested with Adder_Width = 16.

Area Comparison

We can see in Figure that the Hybrid Adder reports a bigger area size, specifically 2646 in total (LUTs + WNS), versus the area of Ripple Carry Adder which is 2671 in total (LUTs + WNS). Therefore, we can include that the area size is not improved when using the Hybrid Adder.

Name 1	Slice LUTs (53200)	Slice Registers (106400)	Bonded IOB (125)	BUFGCTRL (32)
N mp_adder	1073	1573	1541	1

Figure 11: The Post-Synthesis Utilization report of the ripple carry adder from LAB#3 tested with Adder_Width = 16.

Name 1	Slice LUTs (53200)	Slice Registers (106400)	Bonded IOB (125)	BUFGCTRL (32)
N mp_adder	1099	1572	1541	1

Figure 12: The Post-Synthesis Utilization report of the hybrid adder tested with Adder_Width = 16

We can see that the obtained results are in line with the expectations mentioned previously in the Technical description section. When the Adder_Width is further extended to 32 bits, the number of LUTs increases, but not the Slice Registers, as seen in Figure 7. Nonetheless, there was still a significant total increase in area (1099+1572 = 2671 for Adder_Width = 16 and 1198+1570=2768 for Adder_Width = 32).